

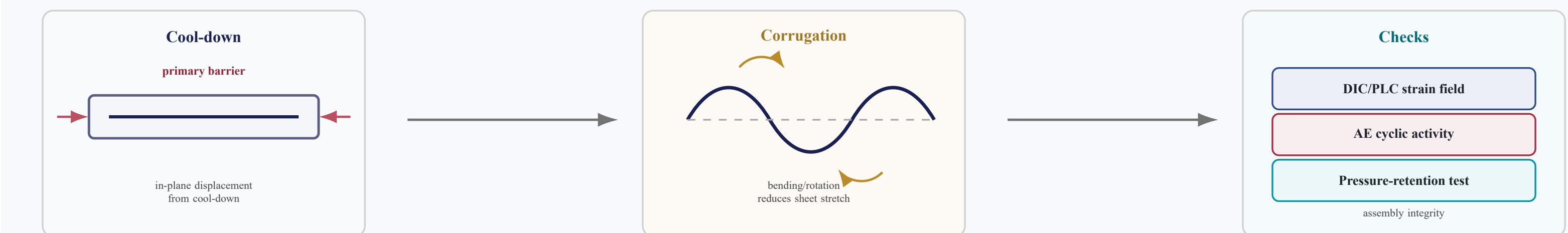
Introduction

Large-scale LH₂ cargo containment needs a metallic primary barrier that is leak-tight, formable, and tolerant of the in-plane displacement generated when the tank system cools from ambient temperature to cryogenic service. If this displacement is carried mainly by membrane stretching, local strain concentration can become a direct risk for cracking, leakage, and loss of containment reliability.

A corrugated 316L membrane is introduced to turn this constraint into a geometric compliance problem. The selected cross-corrugated profile is expected to accommodate imposed displacement through local bending and rotation of crests, valleys, and side-wall transitions, thereby lowering the uniform stretching demand on the thin sheet.

The remaining question is experimental rather than purely geometric: after patent-based forming, specimen preparation, and cyclic loading, does the membrane still follow the intended strain-release mechanism, and can an assembled module retain preliminary integrity after cryogenic cycling?

This poster therefore links three checks into one validation chain: DIC/PLC maps where imposed displacement is released, AE monitors cyclic damage activity during repeated loading, and pressure-hold plus helium leak testing are used as the pressure-retention method to verify assembly integrity after LN₂ fill/drain cycles.



Conceptual schematic: cryogenic contraction sets the displacement demand; corrugation supplies the intended strain-release mechanism; pressure-retention testing checks assembled membrane integrity after LN₂ cycling.

M. Methods: Design-to-Test Logic Chain

Starting point	Value	Why it matters
Selected profile	$\epsilon_p = 1.0, \epsilon_s = 5.0, \lambda = 0.051$	best-performing geometry from the previous design screen
Sheet and material	316L, $t = 1.2$ mm	thin monolithic primary-barrier candidate
Patent basis	CN118237450A	coupled downward pressing and lateral squeezing for equal-section cross corrugation

1. Profile
previous best

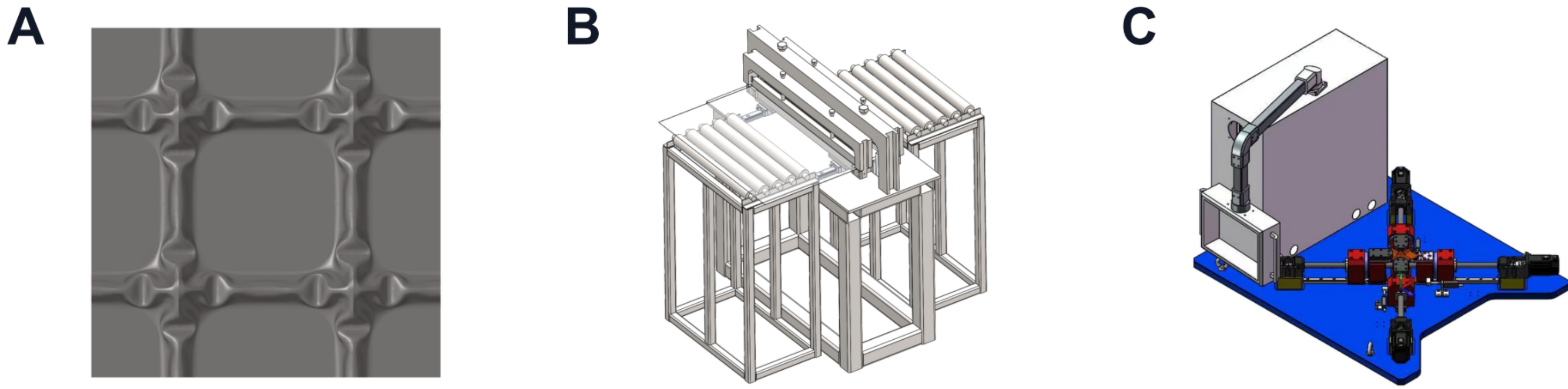
2. Patent process
die + side extrusion

3. Samples
form + cut

6. Assembly
LN₂ cycles

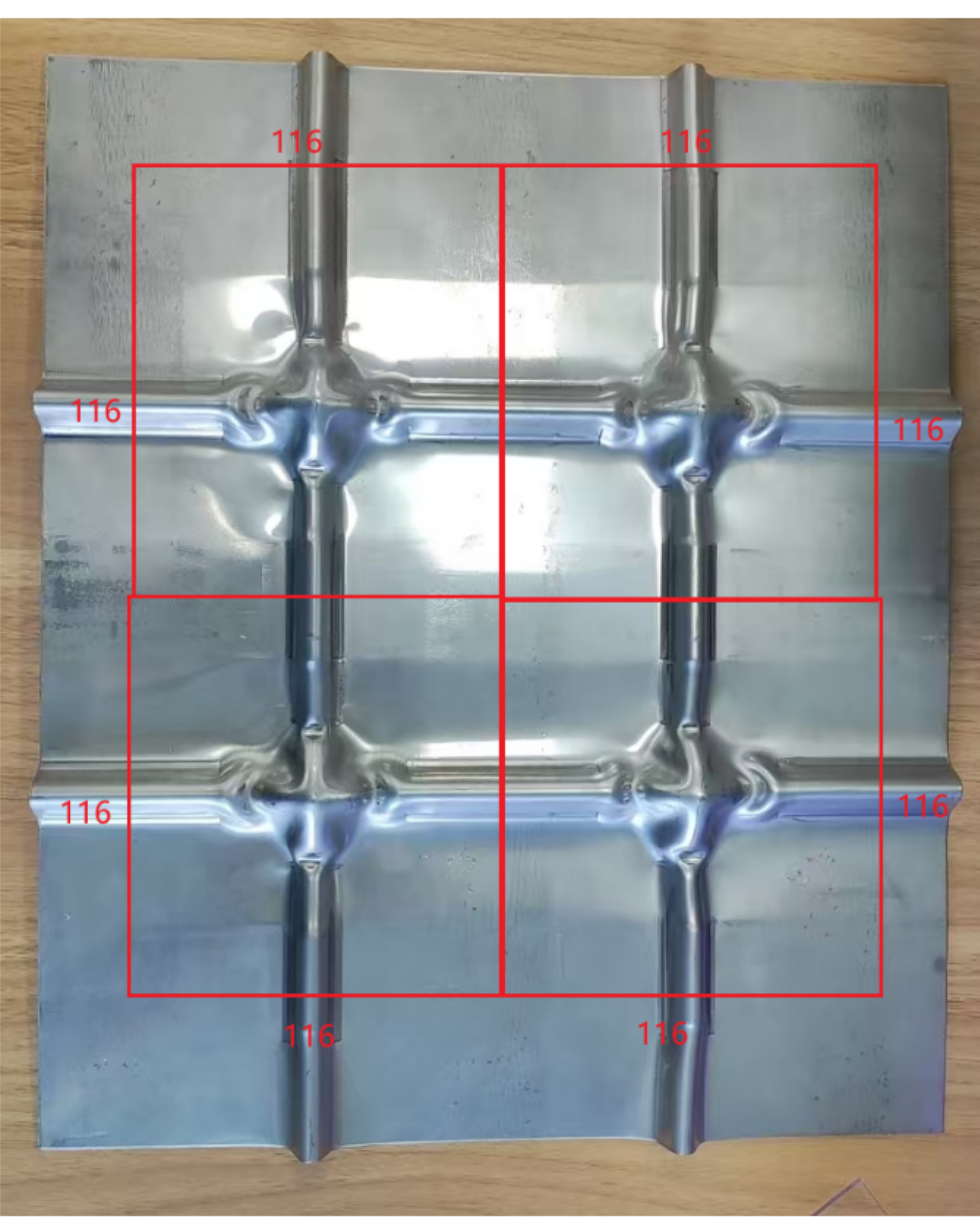
5. AE next
damage evolution

4. DIC first
strain release

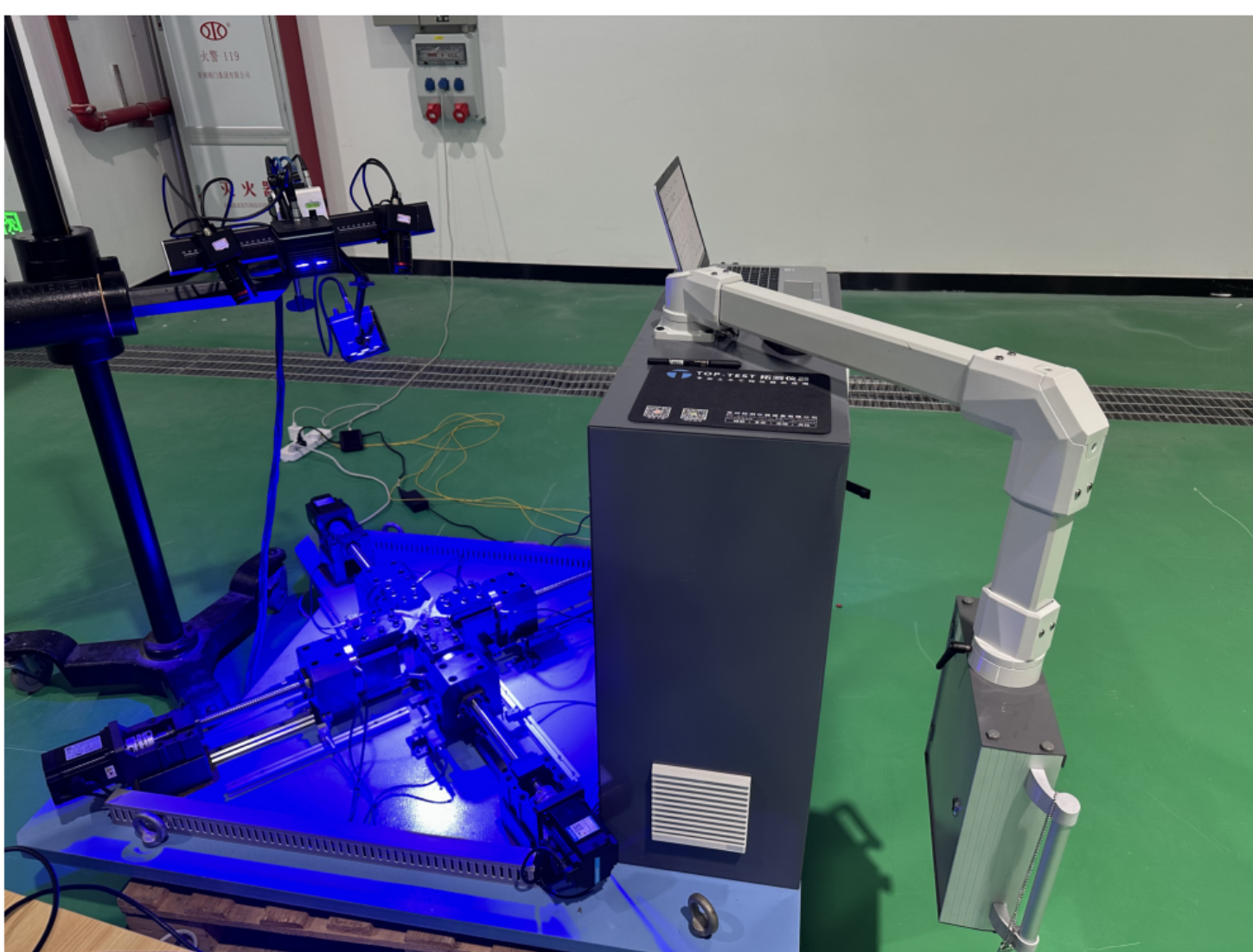


Selected profile → patented processing machine → four-axis specimen platform. The process figure is used as design and preparation context, not as a dimensional CAD claim.

A. Corrugated-panel specimen layout



B. Four-axis DIC/AE test arrangement



Specimen extraction, DIC field of view, and mechanical setup connect the formed profile to the DIC/AE tests.

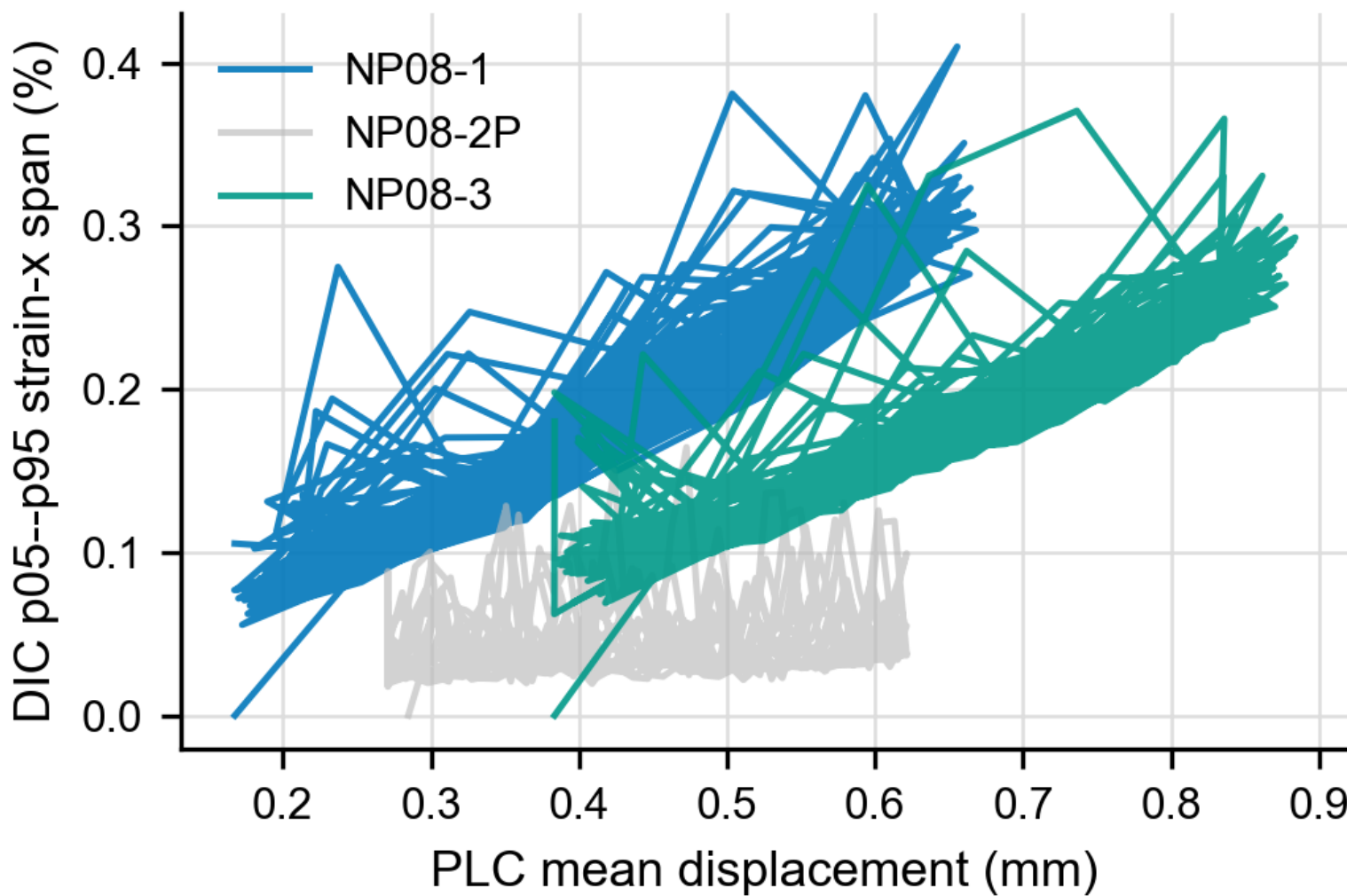
Experiment	Direct question	Measured output
DIC/PLC	How does corrugation release imposed strain?	synchronized strain span, geometry-fixed ROI hierarchy
AE	When does cyclic activity concentrate?	relabeled cycle windows and damage-evolution screen
LN ₂ assembly	Does pressure-retention testing verify assembly integrity after thermal cycling?	pressure-hold and helium leak checks across five LN ₂ cycles

$$\bar{u}_{ik} = \frac{1}{N} \sum_{n=1}^N u_{ik,n}, \quad S_i = \frac{d^2 \bar{u}_i(\bar{x}) / d\bar{x}^2}{d\bar{x}}, \quad \Psi_i = A_{\text{AE}}^2$$

The evidence chain is deliberately ordered: profile and processing first, DIC mechanism second, AE damage-evolution screening third, and pressure-retention testing for assembly integrity last.

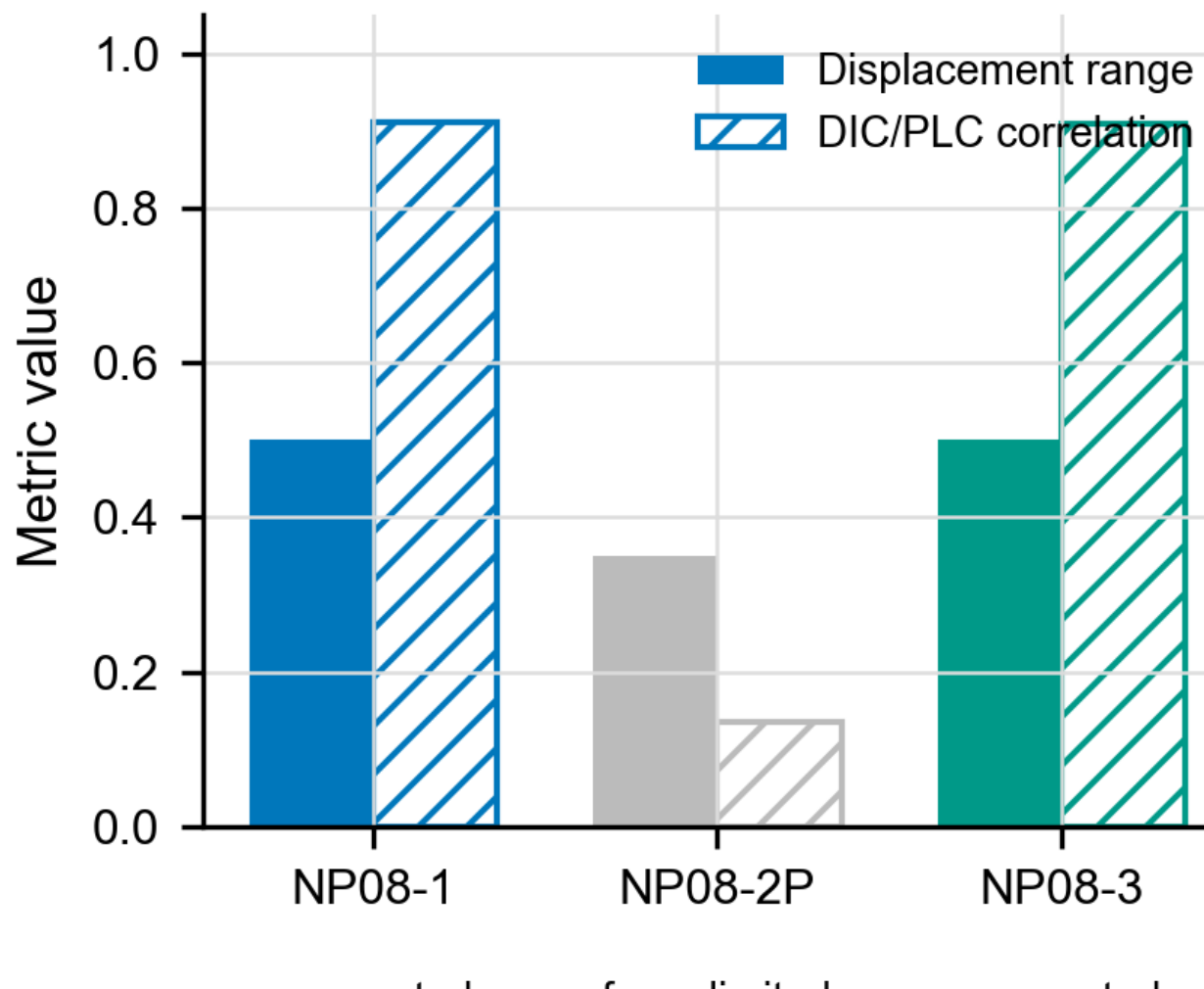
R1. Results: DIC/PLC Mechanism Evidence

A. DIC span synchronized to PLC displacement

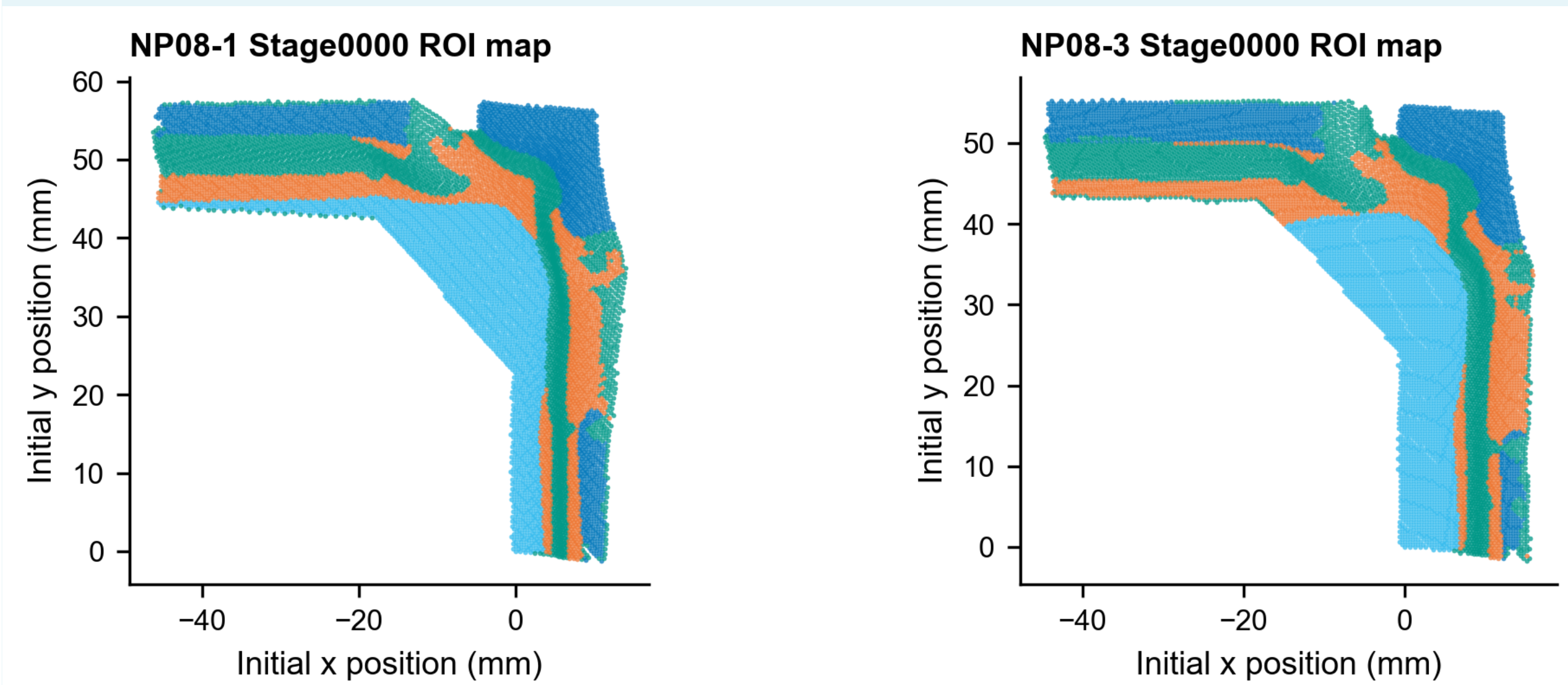


span follows imposed PLC mean displacement.

B. Traceability metrics



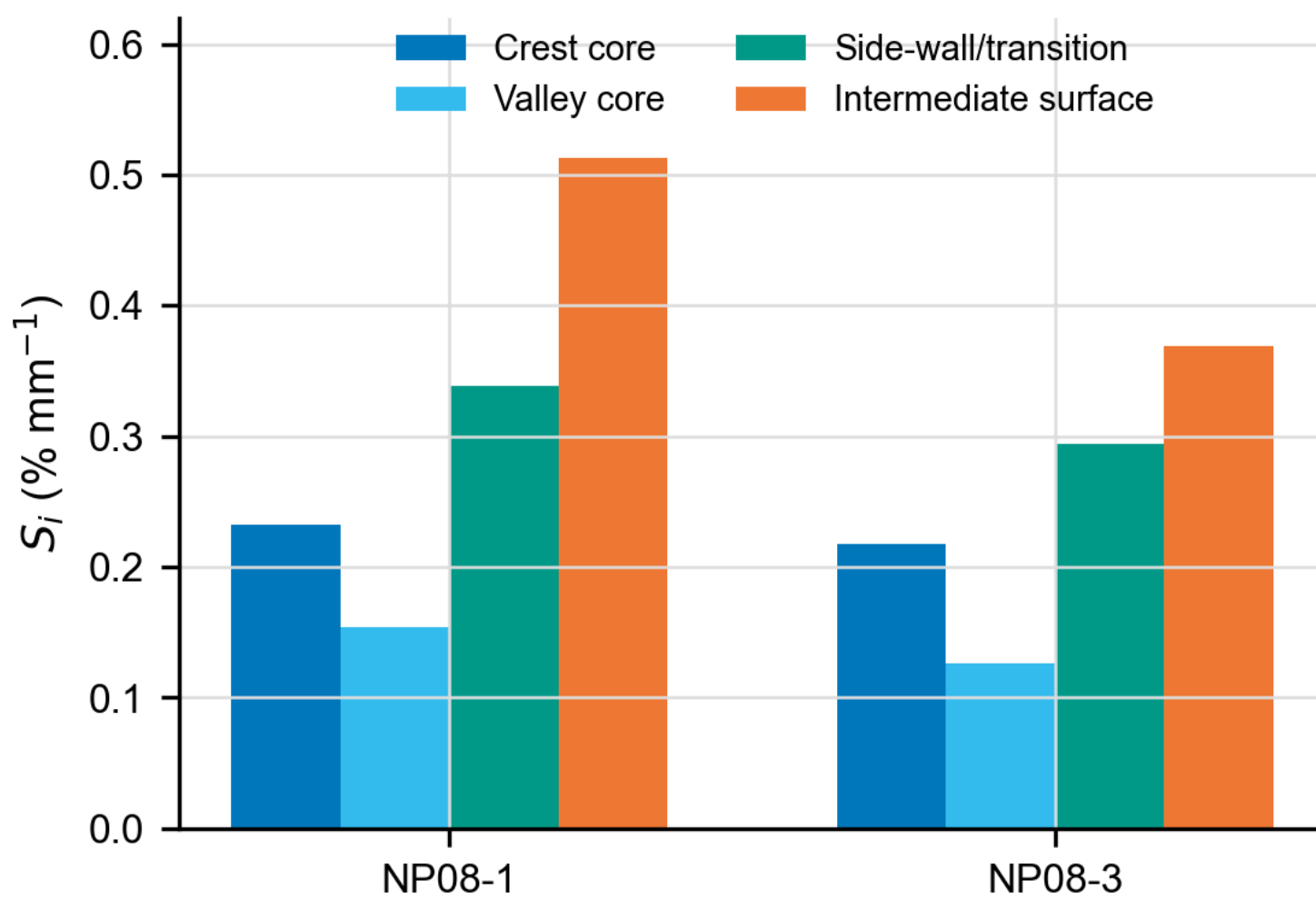
DIC strain



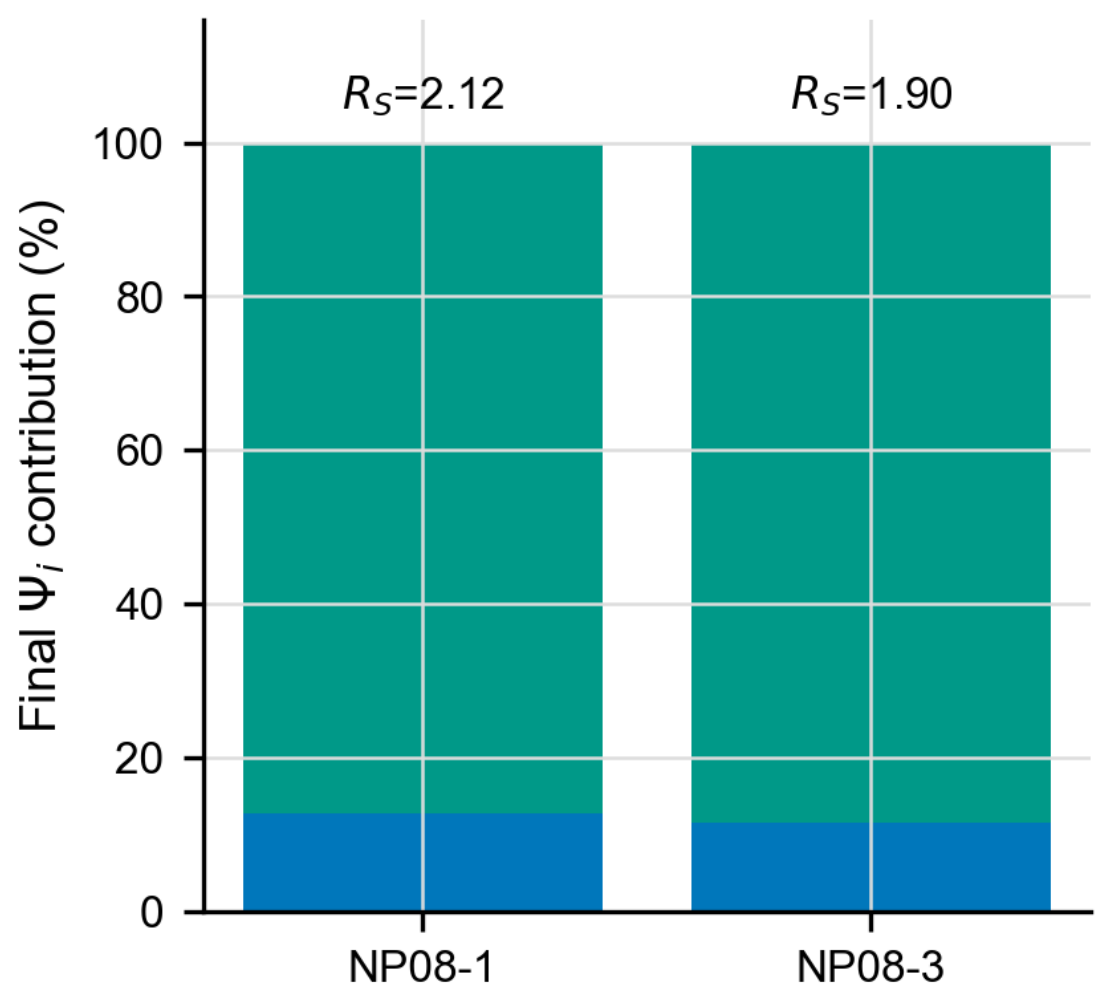
labels.

Initial geometry defines persistent ROI

A. Displacement-normalized ROI strain sensitivity



B. Final strain-demand proxy partition

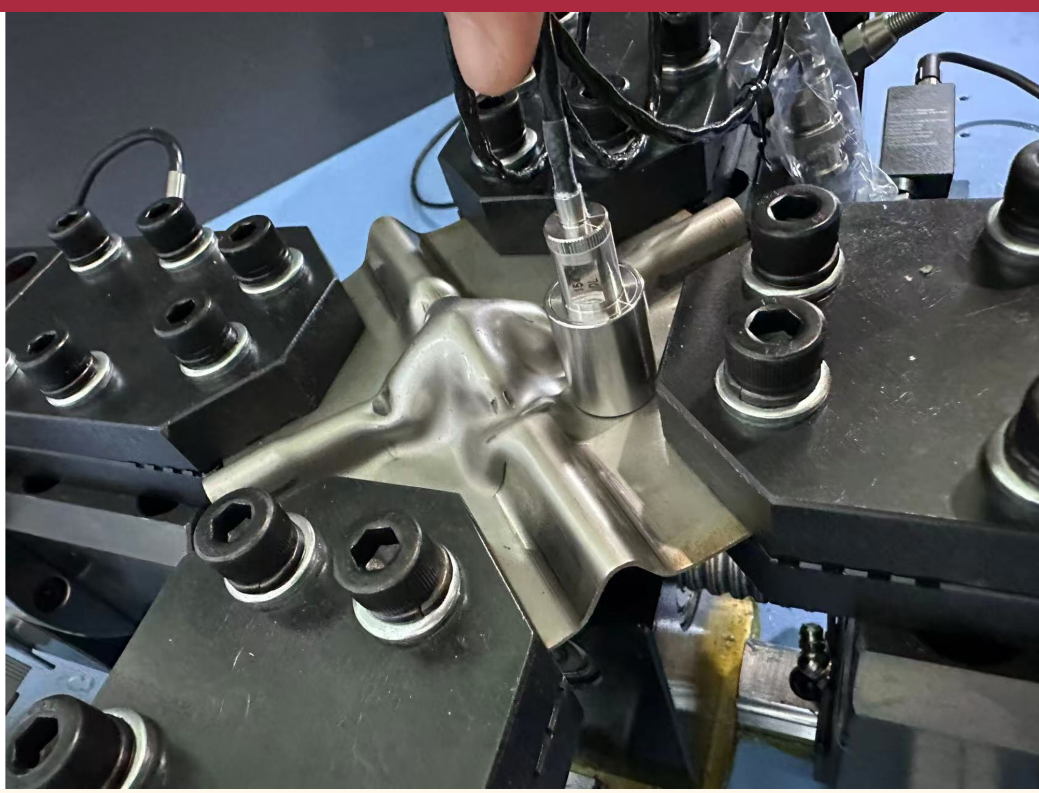


High-deformation ROIs carry most final

Metric	NP08-1	NP08-3	Interpretation
Sensitivity ratio R_S	2.117	1.896	high ROI response is about 2x core response
High-deformation area	54.28%	53.96%	deformation path occupies about half the map
Final strain-demand share	87.2%	88.4%	most demand stays in compliant regions

Mechanism result. Corrugated specimens show repeated ROI hierarchy: side-wall/transition and intermediate surfaces absorb the imposed displacement while crest/valley cores remain lower-demand regions.

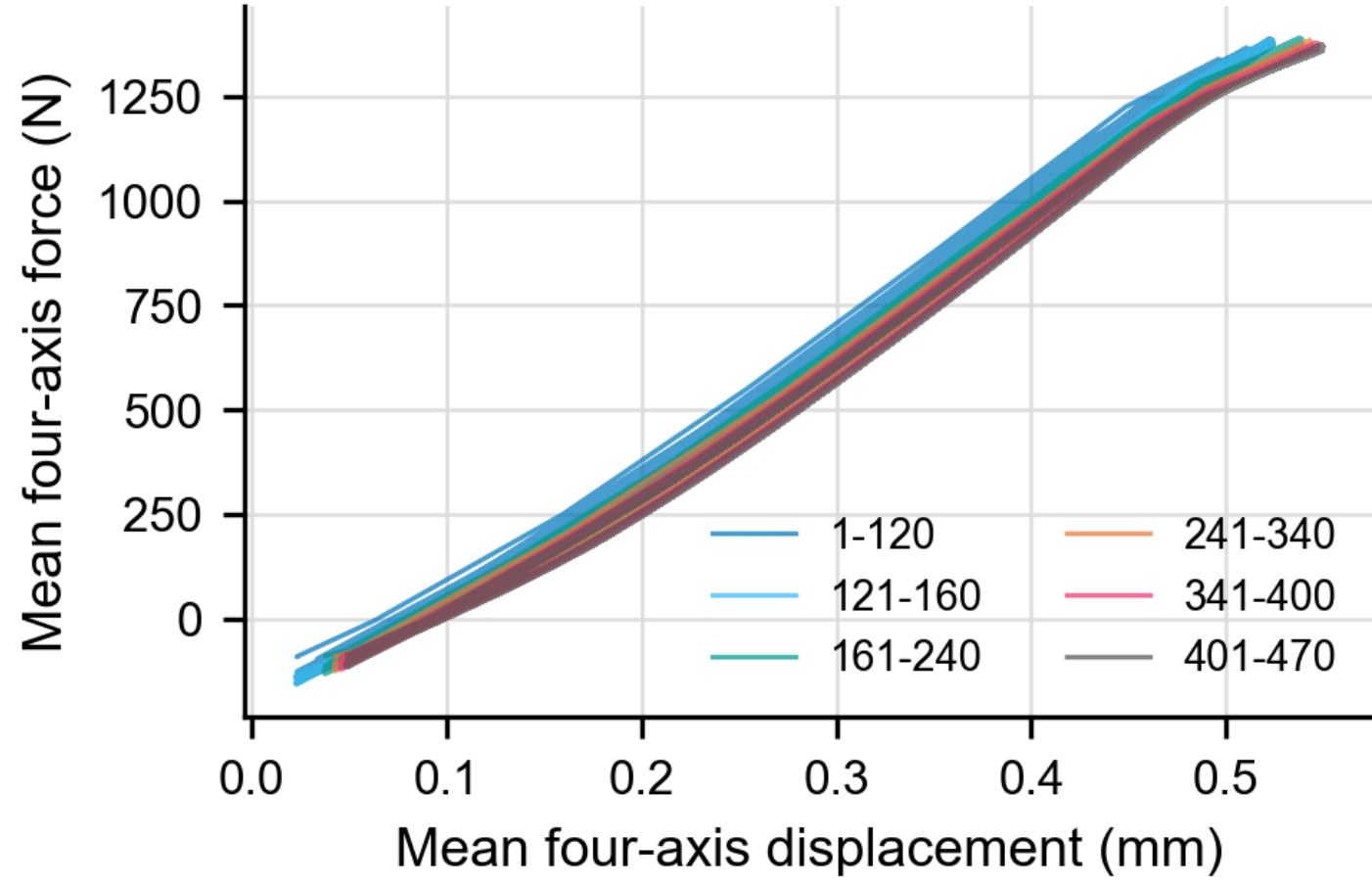
R2. Results: AE Activity and Assembly Integrity Checks



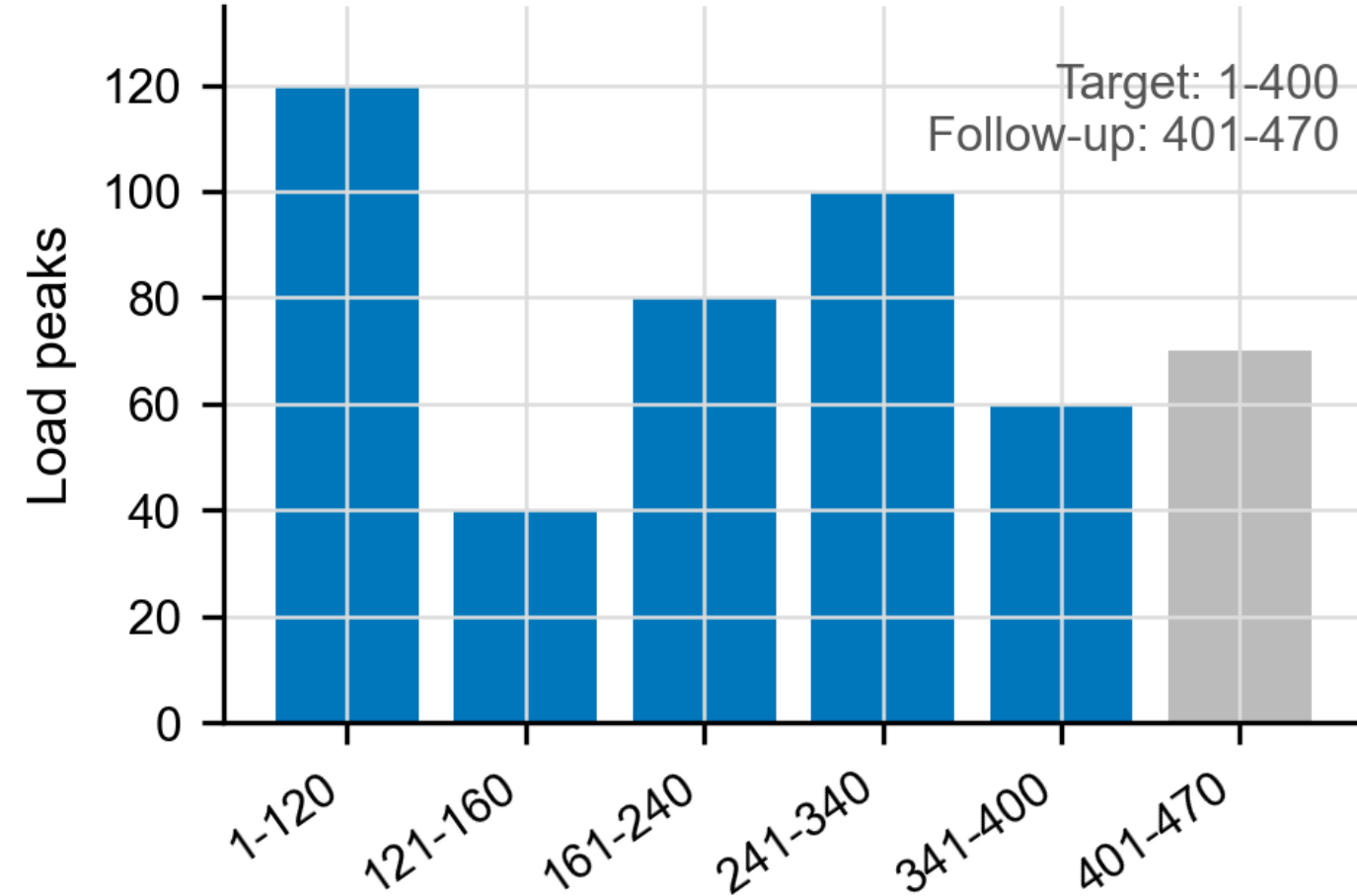
AE sensors were mounted during the same four-axis loading context.

Rule	Interpretive use
Relabel cycles	PLC peak counting resets the AE coordinate before interpretation.
Track features	amplitude, high-amplitude fraction, and energy indicate cyclic activity.
Keep boundary	feature trends screen damage evolution; they do not alone prove cracks.

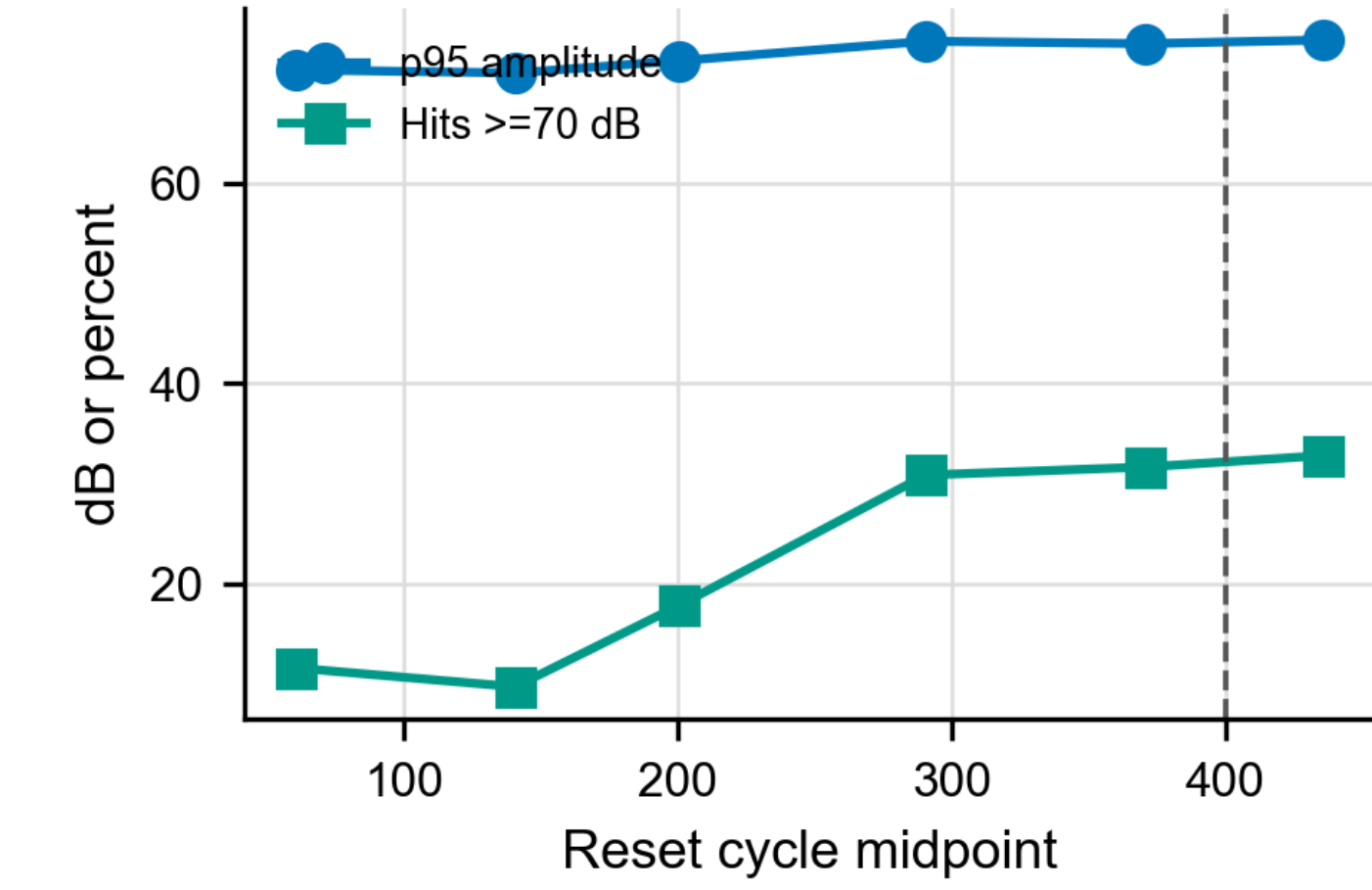
A. PLC force-displacement envelopes



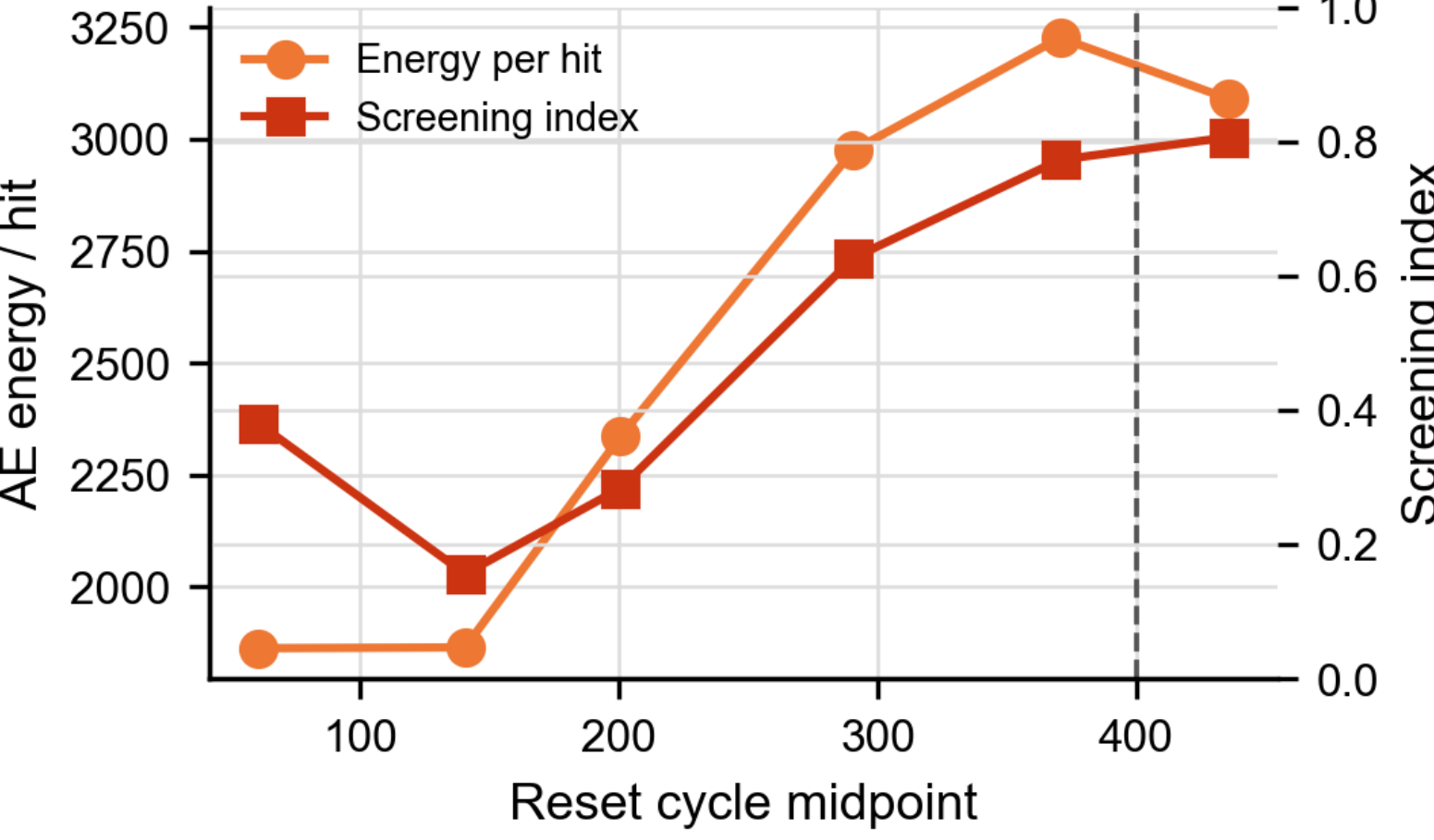
B. PLC-confirmed cycle blocks



C. AE amplitude trend after relabeling

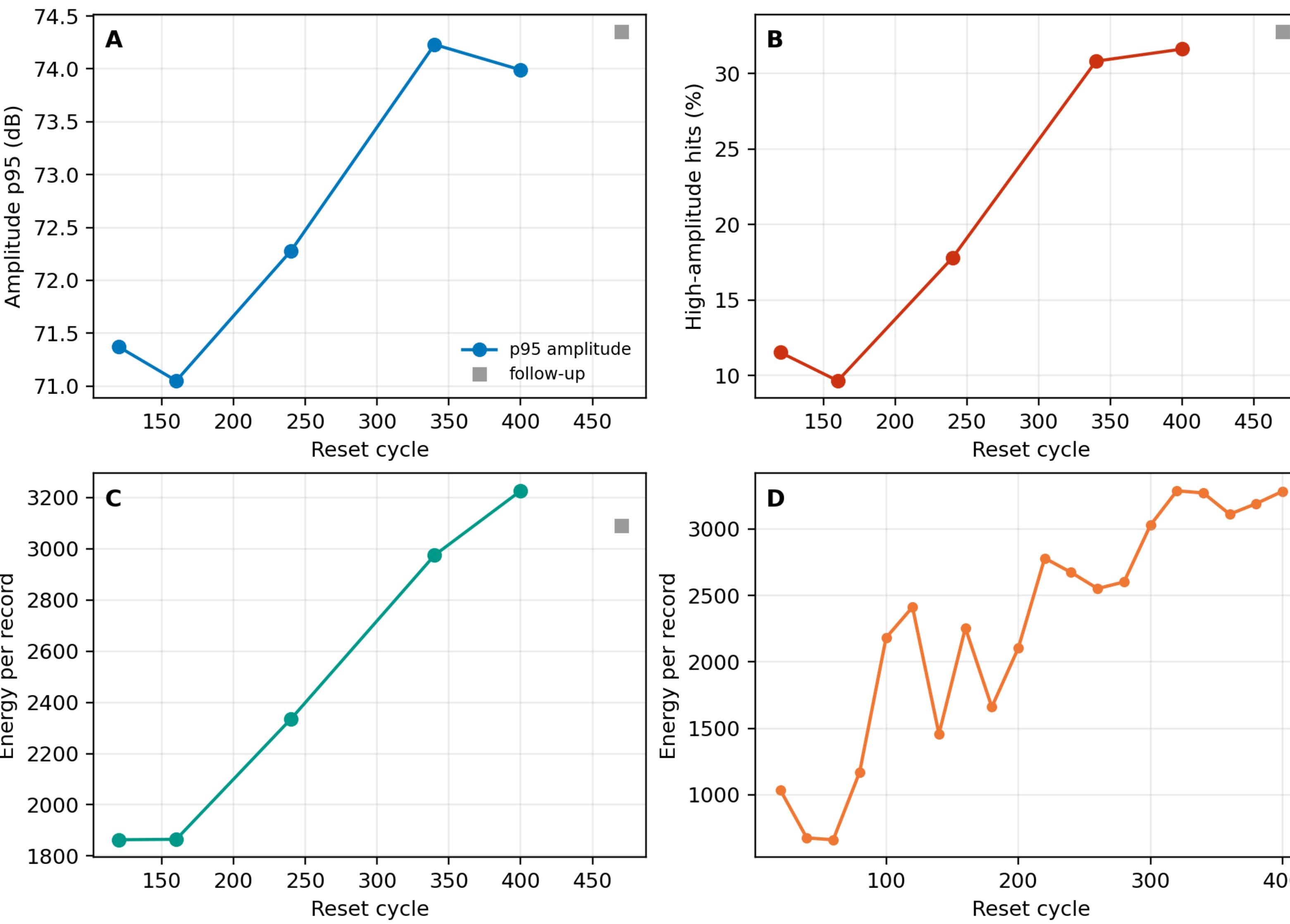


D. AE energy and screening trend



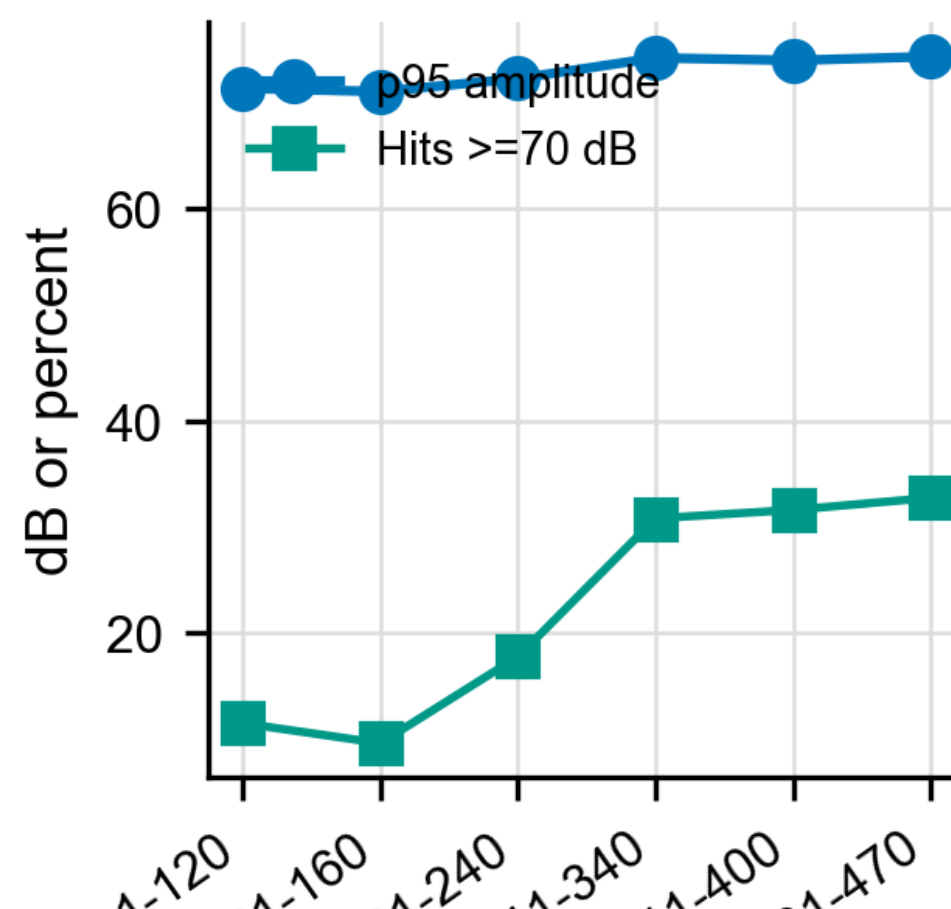
PLC peak counting defines reset cycles 1-400 and follow-up 401-470.

Feature-level AE diagnostics after PLC cycle relabeling

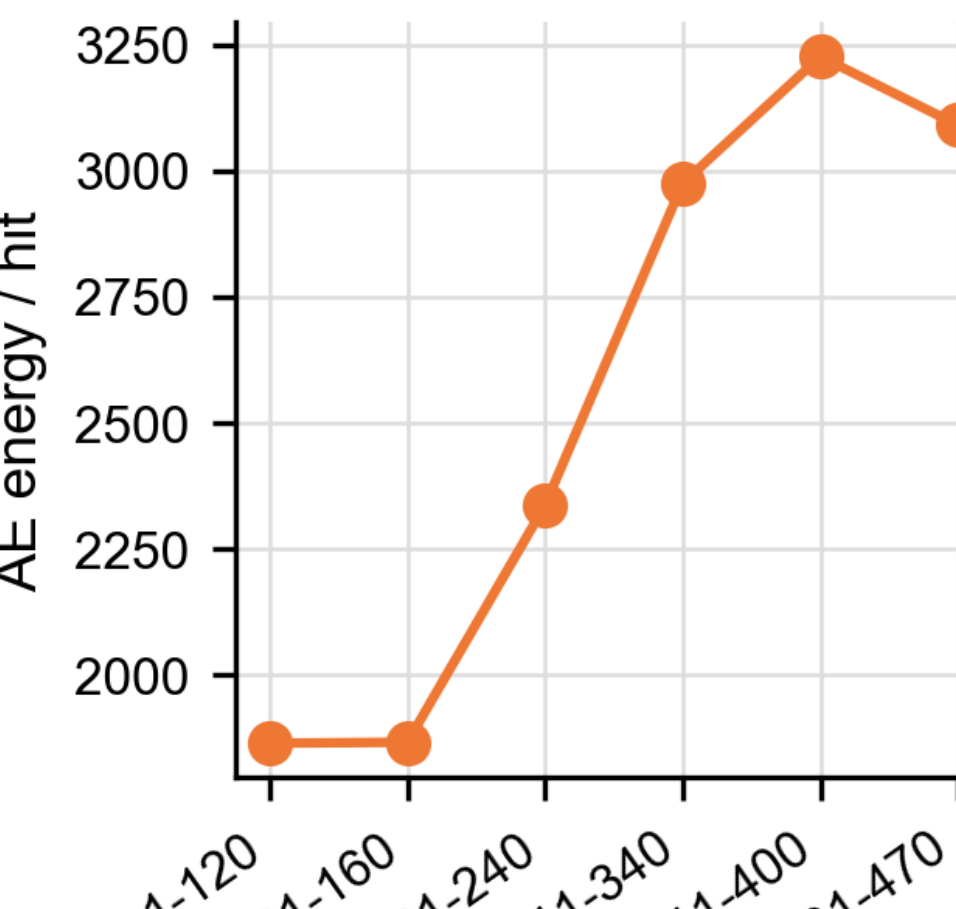


Damage-evolution screen: p95 amplitude, high-amplitude fraction, and energy increase toward the late reset-cycle window.

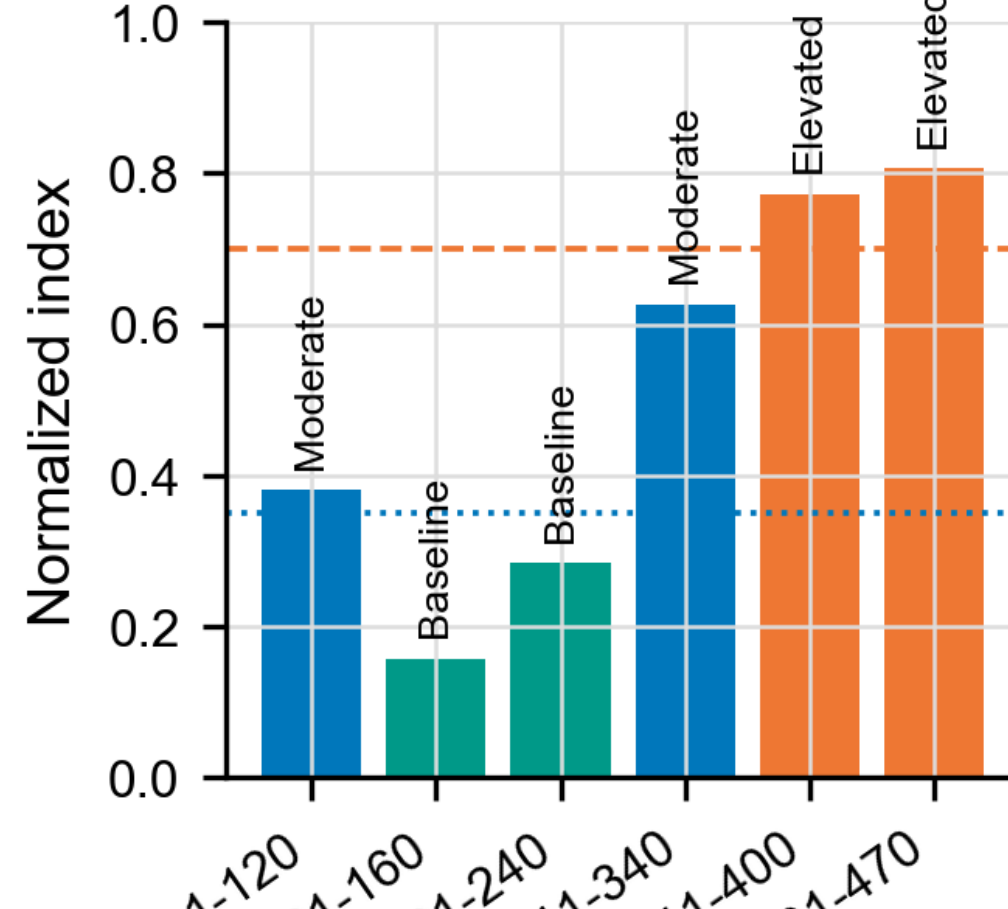
A. High-amplitude activity



B. Energy per hit



C. Screening index



Screening index ranks cycles 341-400 as the inspection-priority windows.

31.63%

records above 70 dB

3.23×10^3

energy per hit

125 kHz

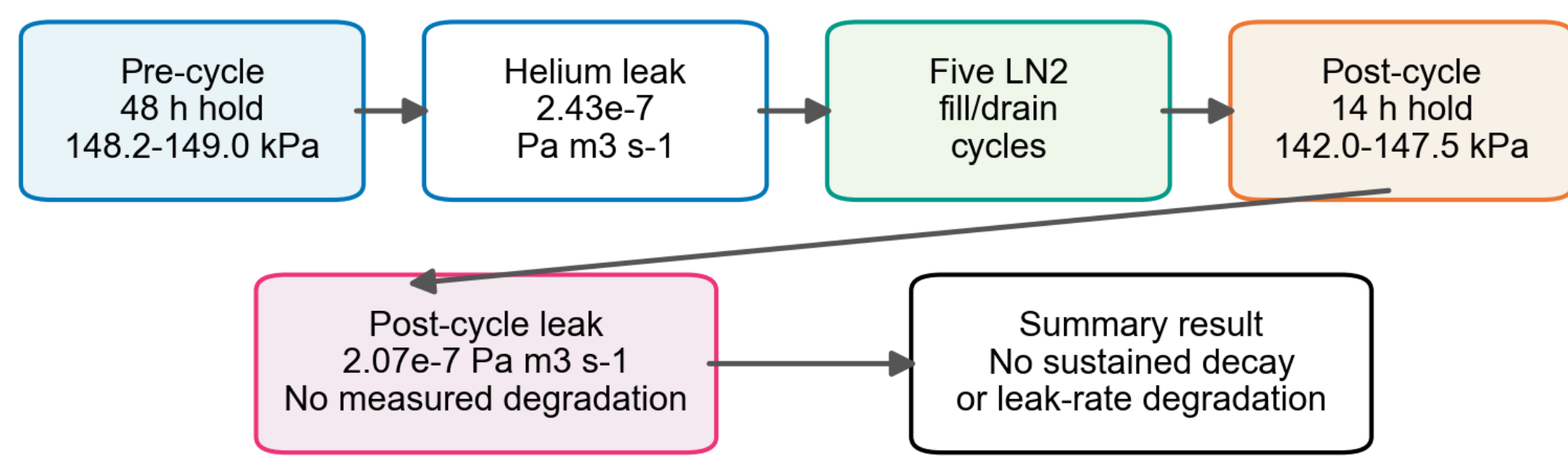
median frequency

1.17M

feature records

Scaled-prototype LN₂ assembly-integrity check

Author-held project summary: 0.2 m3 integrated membrane, pressure-retention testing, helium leak detection, and five LN₂ filling/drain cycles.



Claim boundary: pressure-retention is the test method; the supported objective is preliminary assembly integrity, not LH₂ qualification.

Pressure-hold and helium leak-rate checks are the test method; the objective is to verify assembled-membrane integrity before and after five LN₂ filling/drain cycles.

Check	Result
Pre-cycle hold	48 h at 148.2-149.0 kPa abs.
Post-cycle hold	14 h at 142.0-147.5 kPa abs.
Pre-cycle leak	2.43×10^{-7} Pa m ³ s ⁻¹
Post-cycle leak	2.07×10^{-7} Pa m ³ s ⁻¹

D. Discussion: What the Evidence Supports

Layer	Supports	Does not claim	Evidence panel
DIC/PLC	corrugation-driven strain redistribution	service lifetime	R1
AE	late-cycle inspection-priority window	crack growth proof	R2-AE
LN ₂	preliminary assembly integrity	LH ₂ qualification	R2-LN ₂

The strongest evidence is mechanism evidence from synchronized DIC/PLC and ROI mapping.
AE is useful after relabeling because it ranks when inspection attention should focus.
LN₂ pressure-retention testing is necessary but not sufficient for final LH₂ service readiness.

C. Conclusions and Next Gates

Conclusions

- Corrugation redistributes strain into traceable geometry-defined ROIs.
- The late AE reset window gives an inspection-priority signal.
- Pre/post LN₂ pressure and leak checks support preliminary assembly integrity.

Next gates

- Repeat coupon tests with ROI uncertainty.
- Add cryogenic DIC and waveform-level AE.
- Report full acceptance pressure/leak records under LH₂-relevant criteria.

ICEC/ICMC record P1-061, Poster Session I, Exhibition Hall (111+112), June 23, 2026, 14:00-16:00

Claim boundary The evidence supports staged validation under the tested conditions; completed LH₂ service qualification remains outside the present scope.